



Single-cell transcriptome analysis reveals regulatory programs associated with tumor resistance during immunotherapy in colorectal cancer

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Background: Colorectal cancer (CRC), a predominant cancer type globally, is one of the most common cancers worldwide. Immune-checkpoint inhibitors have robust efficacy in the treatment of patients with metastatic DNA mismatch repair (dMMR) CRC; however, some of the patients among them demonstrate resistance to immunotherapy, and the underlying molecular mechanisms remain elusive.

Methods: Using single-cell sequencing data, we identified and annotated cell populations, systematically comparing cell proportions and gene expression changes across different groups. Gene set enrichment analysis was conducted to identify significantly enriched signaling pathways in T-cell subpopulations. Furthermore, we assessed the impact of transcriptional dysregulation in stress response stated T (TSTR) and $\gamma\delta$ T cells on tumor immunotherapy responses and drug resistance. The potential role of T-cell subset interactions in modulating drug resistance and sensitivity was investigated through comprehensive cell communication analysis. Lastly, clinical samples from 190 CRC patients were collected, with paired adjacent normal tissues. Immunohistochemistry was performed using antibodies against FOS and KLRB1.

Results: T cells were identified as the predominant cell population influencing immunotherapy outcomes. T-cell subsets exhibited distinct functional characteristics, with notable differences in their distribution between resistant and sensitive tumor groups. Specifically, exhausted T cells (Tex), GZMK+ T, TSTR, regulatory T cells (Treg), and $\gamma\delta$ T cells were associated with therapeutic resistance. For Tex and GZMK+ T cells, resistance was correlated with the activation of antigen processing and presentation pathways, whereas oxidative stress pathways were downregulated. In contrast, $\gamma\delta$ T cells in the sensitive group exhibited the activation of protein folding pathways, which may contribute to anti-tumor immune responses. Transcriptional network dysregulation in TSTR and $\gamma\delta$ T cells was observed in the drug-resistant group. Cell-to-cell communication analysis showed stronger interactions among T-cell subpopulations, with changes in key signaling pathways linked to treatment resistance. Additionally, downregulation of the CD69-KLRB1 signaling pathway was identified as a potential mechanism of drug resistance in CRC. Lastly, high expression of FOS was significantly associated with a worse prognosis, whereas high expression of KLRB1 predicted improved clinical outcomes, including prolonged overall survival and progression-free survival, and emerged as an independent prognostic factor for CRC patients.

Conclusion: This study highlights the pivotal role of T-cell subsets in patients with metastatic dMMR CRC who resistant to anti-PD-1 therapy, revealing that transcriptional dysregulation and impaired cell communication networks are central mechanisms underlying drug resistance. Notably, KLRB1 has been identified as a promising biomarker for immunotherapy response in CRC patients.

Keywords: anti-PD-1, colorectal cancer, immunotherapy, mechanisms, resistance, T cell

Introduction

Colorectal cancer (CRC) ranks as the third most prevalent cancer worldwide, with more than 1.9 million new cases and 900 000 deaths annually. It is the second most common cause of cancer death, resulting in a significant global disease burden^[1,2]. Although advancements in diagnosis and treatment strategies for CRC have been significant in recent years, many patients continue to face a poor prognosis^[3]. Genomic instability is a critical feature of

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CRC, and the pathogenic mechanisms leading to this condition are categorized into three types: chromosome instability, microsatellite instability, and CpG island methylation phenotype^[4]. Current first-line treatment strategies for CRC primarily adopt a multimodal approach based on tumor-related characteristics. In addition to traditional chemotherapy, immunotherapy has also demonstrated significant clinical efficacy. However, it is important to note that in metastatic CRC, only patients with high microsatellite instability (MSI-H) or deficient mismatch repair (dMMR) derive long-term benefit from immune checkpoint inhibitors^[5]. Immunotherapies, such as anti-PD-1/PD-L1 agents, have demonstrated considerable efficacy in the treatment of metastatic CRC characterized by dMMR/MSI-H mCRC^[6]. The PD-1 receptor contributes to immunosuppression in CRC by interacting with its ligand PD-L1. This interaction activates the downstream signaling pathways in T cells, thereby leading to the suppression of proliferation of T cells, production and release of cytokines, and cytotoxic activity. Such immunosuppression can prevent the development of autoimmune diseases and chronic infections; however, tumor cells may also exploit this mechanism to evade immune surveillance and attack^[7].

Numerous studies have demonstrated that anti-PD-1/PD-L1 treatments enhance CD8+ T cell-mediated anti-tumor immunity^[8,9]. However, many CRC patients also exhibit resistance to immunotherapy, which may be attributed to the dysregulated differentiation of immunosuppressive cells such as regulatory T cells (Treg), myeloid-derived suppressor cells (MDSCs), and tumor-associated macrophages, as well as the upregulation of immunosuppressive proteins in the heterogeneous tumor microenvironment.

These proteins typically induce an anti-inflammatory microenvironment that results in CD8+ T-cell dysfunction and immune resistance^[10]. Therefore, investigating the tumor microenvironment during the development of CRC and elucidating the cellular and molecular mechanisms underlying resistance to CRC immunotherapy will be of crucial importance. This will contribute to the development of various immunotherapy agents and the identification of novel therapeutic targets.

Single-cell transcriptome sequencing (scRNA-seq) allows for the amplification and sequencing of the entire transcriptome at the single-cell level. This technique provides the comprehensive gene expression profiles of individual cells, accurately characterizes cellular heterogeneity^[11–13], elucidates the diversity of immune cell populations within tissues^[14], and facilitates the construction of interaction networks among different cell populations^[12]. The scRNA-seq method provides comprehensive insights that facilitate the understanding of distinct gene expression profiles of individual cells during CRC immunotherapy. By performing scRNA-seq on 18 CRC tumor and non-tumor samples, Wang *et al* identified notable alterations in the composition and functional states of T cells and B cells in tumor tissues. The study revealed that the interaction between circulating B cells expressing CCL8 and CCR5+ T cells may serve as a potential anti-tumor mechanism in advanced CRC tumors^[15]. Li, J. *et al* conducted scRNA-seq analysis on 19 patients with dMMR/MSI-H CRC who received PD-1 blockade therapy. They observed that dynamic changes in the toxicity and proliferation programs of CD8+ T cells were associated with treatment response, while CD4+ Treg were reduced in patients exhibiting a complete response. Moreover, an increased proportion of CD40 B cells was noted^[16]. Wu *et al* utilized scRNA-seq to

HIGHLIGHTS

- Using single-cell sequencing, we identified and annotated distinct T-cell populations, highlighting significant differences in their proportions and gene expression profiles between tumor-sensitive and -resistant groups.
- T cells were found to be the most abundant cell type, with key subsets such as Tex, GZMK+ T, TSTR, Treg, and $\gamma\delta$ T cells playing critical roles in therapeutic resistance.
- The study further explored functional pathways, showing that antigen processing and oxidative stress pathways are linked to resistance, while protein folding pathways activated by $\gamma\delta$ T cells are associated with tumor sensitivity.
- The dysregulation of transcriptional networks in TSTR and $\gamma\delta$ T cells was identified as a key factor driving drug resistance.
- Enhanced cell communication within T-cell subpopulations in resistant groups underscores the importance of intercellular interactions in treatment outcomes.

characterize differences in the tumor microenvironment of dMMR/MSI-H mCRC between anti-PD-1 treatment-sensitive and treatment-resistant groups. Their findings revealed that CD8+ T cells and IL-1 β were the cell types and genes most strongly associated with drug resistance, respectively. Furthermore, the IL-1 β -induced infiltration of MDSCs was identified as a key contributor to resistance in CRC^[17].

These single-cell transcriptomic datasets provide valuable insights into the cellular and molecular mechanisms underlying CRC immunotherapy and serve as a critical resource for the development of novel targeted therapeutic strategies. Based on these findings, we employed bioinformatics methods to identify specific cell populations and abnormal regulatory patterns that may be linked to response or resistance in CRC immunotherapy^[18]. This will deepen our comprehension of the underlying fundamental molecular mechanisms.

Material and methods

Retrieval and process of public data

The scRNA-seq datasets used in this study were obtained from the NCBI repository at <https://www.ncbi.nlm.nih.gov/sra/PRJNA932556>. The unique molecular identifier (UMI) count matrix was converted into a Seurat object using the R package Seurat^[19] (version 4.0.4). Cells exhibiting UMI counts below 500, the number of detected genes was less than 200, or mitochondrial-derived UMI counts exceeding 30% were classified as low-quality and subsequently excluded from further analysis. Additionally, genes detected in fewer than three cells were also excluded from subsequent analysis.

scRNA-seq data preprocessing and quality control

Following the implementation of quality control measures, the UMI count matrix was subjected to log-normalization. Subsequently, the top 2000 variable genes were utilized to generate potential anchors using the FindIntegrationAnchors function in Seurat. The data were then integrated using the

IntegrateData function. Principal component analysis was performed on the integrated data matrix, and the top 50 principal components were selected for subsequent analysis using the Elbowplot function in Seurat. The primary cell clusters were identified using the FindClusters function, with the resolution parameter set to the default value ($res = 0.6$). This process yielded 25 distinct cell clusters, which were visualized using T-distributed stochastic neighbor embedding (tSNE) or uniform manifold approximation and project (UMAP) plots. The cell types within each cluster were identified by detecting gene markers using the “FindMarkers” function in the Seurat package (v4.0.4). Subsequently, cell types were annotated using marker genes from Cell Taxonomy^[20].

Differential gene expression analysis

Differentially expressed genes (DEGs) were identified utilizing the FindMarkers/FindAllMarkers function from the Seurat package, applying a one-tailed Wilcoxon rank sum test with *P*-values adjusted for multiple testing using the Bonferroni correction. DEGs were computed considering all genes with an expression difference on a natural log scale of at least 0.5 and an adjusted *P*-value below 0.05.

Gene set enrichment analysis

We utilized gene set enrichment analysis (GSEA) to evaluate sensitivity and resistance to anti-PD-1 therapy in MSI-H mCRC^[21]. GSEA is a gene set-based analysis method designed to determine whether a predefined set of genes is significantly enriched among DEGs. First, we constructed a gene expression matrix and subsequently analyzed it using gene sets from the Molecular Signatures Database (MSigDB), including Gene Ontology (GO) (msigdbv2023.1.HsGMTs/c5.go.bp.v2023.1.Hs.entrez.gmt) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways (msigdbv2023.1.HsGMTs/c2.cp.kegg.v2023.1.Hs.entrez.gmt). A positive Normalized Enrichment Score (NES) indicates that the gene set is enriched in the resistant group, and a negative NES value indicates enrichment in the sensitive group.

Transcription factor regulatory network analysis

The transcription factor (TF) modules were identified using the SCENIC^[22] Python workflow (version 0.11.2) with default parameters (<http://scenic.aertslab.org>). A comprehensive list of human TF genes was retrieved from the pySCENIC resources (<https://github.com/aertslab/pySCENIC/tree/master/resources>). Activated TFs were identified from the Area under curve (AUC) matrix, and differentially activated TFs were selected using the limma package in R^[23]. To identify cluster-specific regulons, especially in analyses involving multiple cell types where certain regulons may be shared across clusters, we employed the Regulon Specificity Score (RSS)^[24]. The regulatory networks of the modules containing TFs and their corresponding target genes were visualized using Cytoscape (v3.9.1) (<https://cytoscape.org/>).

Pseudotime analysis

Monocle 2 (version 2.26.0) was utilized to perform pseudotime analysis, trajectory construction, and the calculation of

pseudotime-dependent gene expression^[25]. Dimension reduction was performed using DDRTree (version 0.1.5). Cell trajectories were visualized according to cell subtypes and states. The Basic Differential Analysis algorithm was used to identify genes that were differentially expressed along the pseudotime trajectory. Branched Expression Analysis Modeling was employed to identify genes regulated in a branching-dependent manner.

Cell-cell communication

Cell-cell interactions were inferred using CellChat (v1.0.0) based on the expression of known ligand–receptor pairs in different cell types^[26]. CellChat requires gene expression data as input and estimates the likelihood of cell–cell communication by integrating gene expression profiles with prior knowledge of interactions among various signaling ligands, receptors, and cofactors. To identify potential cell–cell communication networks, we strictly followed the officially established workflow. We input the normalized counts into CellChat and applied the following preprocessing functions: identifyOverExpressedGenes, identifyOverExpressedInteractions, and projectData using standard parameters. CellChatDB, a ligand–receptor interaction database, is a manually curated repository that provides literature-supported interactions between ligands and receptors in both humans and mice. For the primary analyses, core functions computeCommunProb, computeCommunProbPathway, and aggregateNet were utilized with standard parameters and fixed randomization seeds. Subsequently, the function netAnalysis_signallingRole was applied to the netP data slot to identify the senders and receivers within the network.

Functional enrichment analysis

Functional categories of genes were determined by identifying GO terms and KEGG pathways using KOBAS 2.0^[27]. The hypergeometric test and Benjamini–Hochberg False discovery rate (FDR) correction were used to assess the enrichment of each term.

Clinical sample collection

A total of 190 CRC tissue samples and their corresponding adjacent normal tissues were obtained from patients who underwent radical resection in our hospital. None of the patients had received any preoperative treatment, and all patients had at least 5 years of follow-up data. All experiments were conducted in accordance with the Declaration of Helsinki, and the study was approved by the Ethics Committee of the Hospital. Tumor recurrence or regression was determined based on carcinoembryonic antigen (CEA) levels and liver-specific magnetic resonance imaging.

Immunohistochemistry analysis

Immunohistochemistry (IHC) staining was conducted following the previously described protocol^[28]. A rabbit polyclonal antibody against FOS (ab222699, Abcam, Cambridge, UK) and KLRB1 (ab234107, Abcam, Cambridge, UK) were used in this study. IHC staining results were scored based on two parameters: staining intensity (graded on a scale of 0–3) and the percentage of positively stained cells (graded on a scale of 1–4). The final IHC score was calculated by multiplying intensity and percentage. The

evaluations were independently conducted by two pathologists who were blinded to the clinical data.

Results

scRNA-seq analysis of tumor tissue from MSI-H mCRC patients undergoing anti-PD-1 monotherapy identified distinct cell populations

Following the completion of preliminary quality control procedures, transcriptome mapping data from 43 335 cells were obtained. Subsequent processing involved selecting the top 50 principal components for UMAP-based dimensionality reduction and visualization, resulting in the identification of 25 distinct cell clusters (clusters 0–24). Based on literature and well-characterized cell markers (Supplementary Digital Content Fig. S1A, available at: <http://links.lww.com/JS9/F57>), we identified 11 major cell types (Fig. 1A), including T cells, B cells, monocytes, dendritic cells, plasma cells, mast cells, epithelial cells, endothelial cells, cup cells, clustered cells, and cancer-associated fibroblasts (CAFs). We subsequently analyzed the cell proportions in both the sensitive and resistant groups. The cellular composition of the two groups was largely comparable, predominantly composed of immune cells, such as T cells (Fig. 1B). Moreover, each cell type exhibited high expression of specific signature marker genes (Fig. 1C), which may play a crucial role in defining cellular identity and function. We also observed a significant increase in the proportion of monocytes, as well as a notable decrease in the proportions of intestinal epithelial and clustered cells in the drug-resistant group (Supplementary Digital Content Fig. S1B, available at: <http://links.lww.com/JS9/F57>). Furthermore, the analysis of gene expression profiles indicated that endothelial cells and monocytes exhibited the highest number of DEGs. Notably, the gene expression profiles of T cells, which constituted the predominant proportion of the samples, remained unaltered (Supplementary Digital Content Fig. S1C, available at: <http://links.lww.com/JS9/F57>). This lack of change is likely attributable to the inherent heterogeneity among T-cell subclusters. Next, we investigated the alterations in gene expression profiles across all clusters. It was evident that specific T-cell subclusters (C2, C3, C4, C9, C10, C14, and C15) exhibited notable alterations in their gene expression profiles (Supplementary Digital Content Fig. S1D, available at: <http://links.lww.com/JS9/F57>).

Subsequently, we performed a GO functional enrichment analysis on the DEGs in each distinct cell cluster separately. The results indicated that, in the drug-resistant group, the genes that were significantly upregulated in the C14:T-cell subcluster were primarily enriched in pathways related to immune response. Conversely, the genes that were significantly upregulated in the C3:T-cell subcluster, although involved in the immune response, were also closely linked to the NF- κ B signaling pathway (Fig. 1D). These findings suggest that different T-cell subclusters may play crucial roles in mediating tumor treatment responses. Similarly, genes that were significantly downregulated in different cellular subclusters were markedly enriched in various functional pathways (Supplementary Digital Content Fig. S1E, available at: <http://links.lww.com/JS9/F57>).

Single-cell analysis revealed the differences in T-cell subsets between sensitivity and resistance to anti-PD-1 therapy in patients with MSI-H mCRC

T cells represent a critical component of the immune response and serve as a major target for anti-tumor therapies. We isolated individual T cells and performed secondary clustering, identifying 19 distinct clusters (Fig. 2A). Initially, based on the intrinsic markers CD4 and CD8A, the cells were categorized into CD4+ T cells, CD8+ T cells, and other T cell types (Supplementary Digital Content Fig. S2A, available at: <http://links.lww.com/JS9/F57>). Moreover, using markers specific to each cluster, the clusters were further categorized into 16 distinct T-cell subpopulations (Fig. 2B). Furthermore, we identified several unconventional and other types of T cells. Unconventional T cells include $\gamma\delta$ T cells (TRDChiTRGC1/2hi), natural killer T cells (NKT, KLRC1hiNCAM1hi), and mucosa-associated invariant T cells (MAIT, KLRB1hiIL23RhiSLC4A10hi). Other T-cell types include proliferating T cells co-expressing STMN1 and TUBB (Proliferating_T), pro-inflammatory T cells expressing inflammatory mediators such as S100A8/A9 (Inflammatory_T), dysfunctional T cells expressing GRAIL (encoded by RNF128), and NEDD4L (Anergic_T), as well as a distinct T-cell clusters characterized by elevated expression of tropomyosin (TTN), referred to as (TTN+ T) (Fig. 2C). Analysis of the proportions of each T-cell subpopulation revealed that the percentage of CD8+ T cells was higher in the sensitive group (29.1%) than in the drug-resistant group (28.49%). This finding supports the conclusion that CD8+ T cells serve as the primary effectors in the anti-tumor immune response. Notably, the proportion of CD4+ T cells in the drug-resistant group (27.35%) was significantly higher than that in the sensitive group (19.13%), indicating a potentially crucial role of CD4+ T cells in anti-tumor immunity (Supplementary Digital Content Fig. S2B, available at: <http://links.lww.com/JS9/F57>). Among the CD8+ T-cell subsets, Tex and granzyme K+ (GZMK+) T cells represent the primary cytotoxic populations that are capable of directly eliminating tumor cells. However, these cells gradually exhibit a reduced proliferative capacity, and their elevated levels in the resistant group may indicate a lack of response to immunotherapy (Fig. 2D). Beyond identity markers, several cell-specific marker genes were identified across different T-cell subpopulations (Fig. 2E). We also evaluated the impact of anti-PD-1 treatment on key immune checkpoint pathways. Our findings demonstrate that PD1 (PDCD1), CTLA4, LAG3, TIM3 (HAVCR2), TIGIT, and BTLA are predominantly expressed in Tex, Tph, Tfh, and Treg cells. Furthermore, an NK cell-associated immune checkpoint, NKG2A (KLRC1), was assessed and found to exhibit high expression levels in NK_T, $\gamma\delta$ T, and MAIT cells. Notably, these immune checkpoints demonstrated distinct expression patterns between the sensitive and resistant groups (Fig. 2F).

Identification of potential functional pathways driving tumor resistance in T-cell subsets

Considering the functional roles of various T-cell subsets in immune responses and integrating previous analyses, we focused on CD8-high-expressing exhausted T-cell populations (including Tex and GZMK+ T cells), TSTR cells, CD4-high-expressing Tregs, and unconventional $\gamma\delta$ T cells. GSEA analysis demonstrated a significant upregulation of pathways related to antigen processing

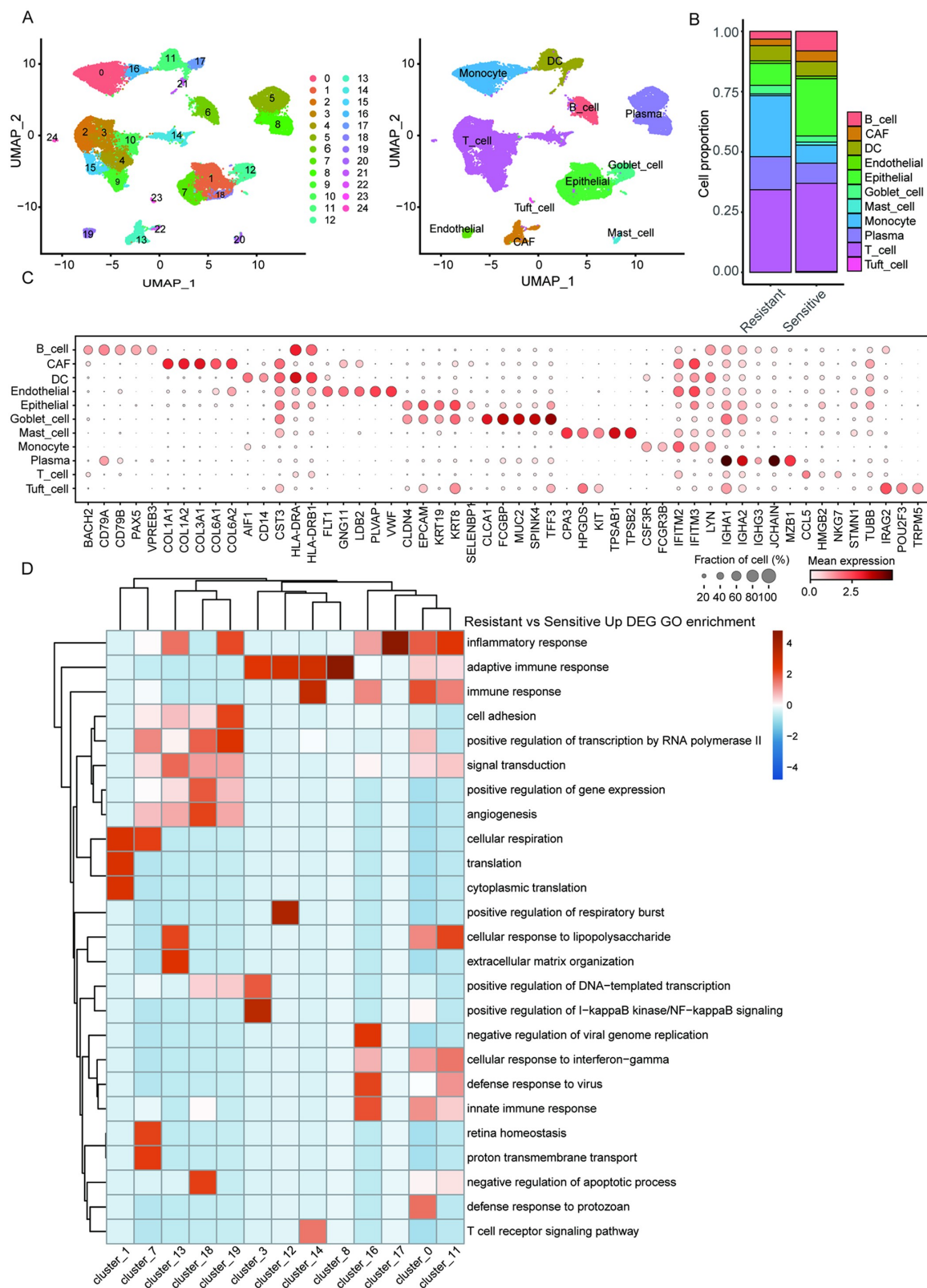


Figure 1. ScRNA-seq analysis of tumor tissue from MSI-H mCRC patients undergoing anti-PD-1 monotherapy. (A) UMAP of 43 335 cells from 6 CRC samples, color-coded according to 11 major cell types identified by signature gene expression profiles. (B) Proportion of 11 cell types in the sensitive and resistant groups. (C) Dot plot illustrating the top five marker genes in each cell type, with darker red indicating higher levels of gene expression. (D) GO biological process enrichment of upregulated genes between the resistant and sensitive groups. scRNA-seq, single-cell transcriptome sequencing; MSI-H, high microsatellite instability; CRC, colorectal cancer; UMAP, uniform manifold approximation and project; GO, gene ontology.

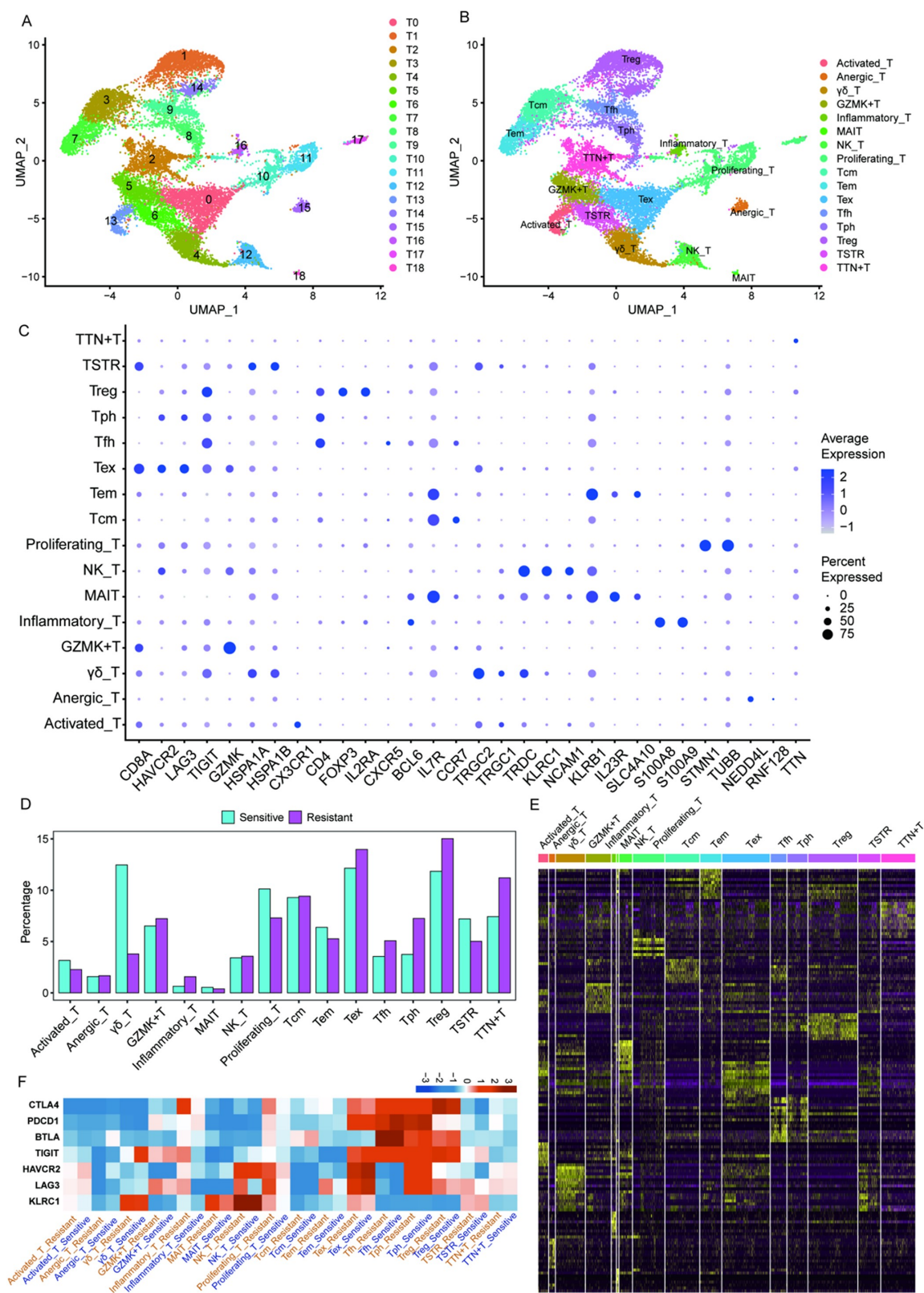


Figure 2. Comparison of T-cell subsets between sensitivity and resistance to anti-PD-1 therapy in patients with MSI-H mCRC. (A) and (B) UMAP visualization of 15,327 T cells from six CRC samples, colored by 19 cell subclusters and annotated into 16 distinct cell subtypes based on marker gene expression. (C) Dot plot depicting expression of representative genes in each T-cell subtype. (D) Bar plot comparing the proportions of cell populations of each T-cell subtype in the respective sample group. (E) Heatmap displaying the expression of the top 10 marker genes of each T-cell subtype. (F) Heatmap comparing the expression levels of seven immune checkpoint genes between the resistant and sensitive sample groups. MSI-H, high microsatellite instability; UMAP, uniform manifold approximation and project; CRC, colorectal cancer.

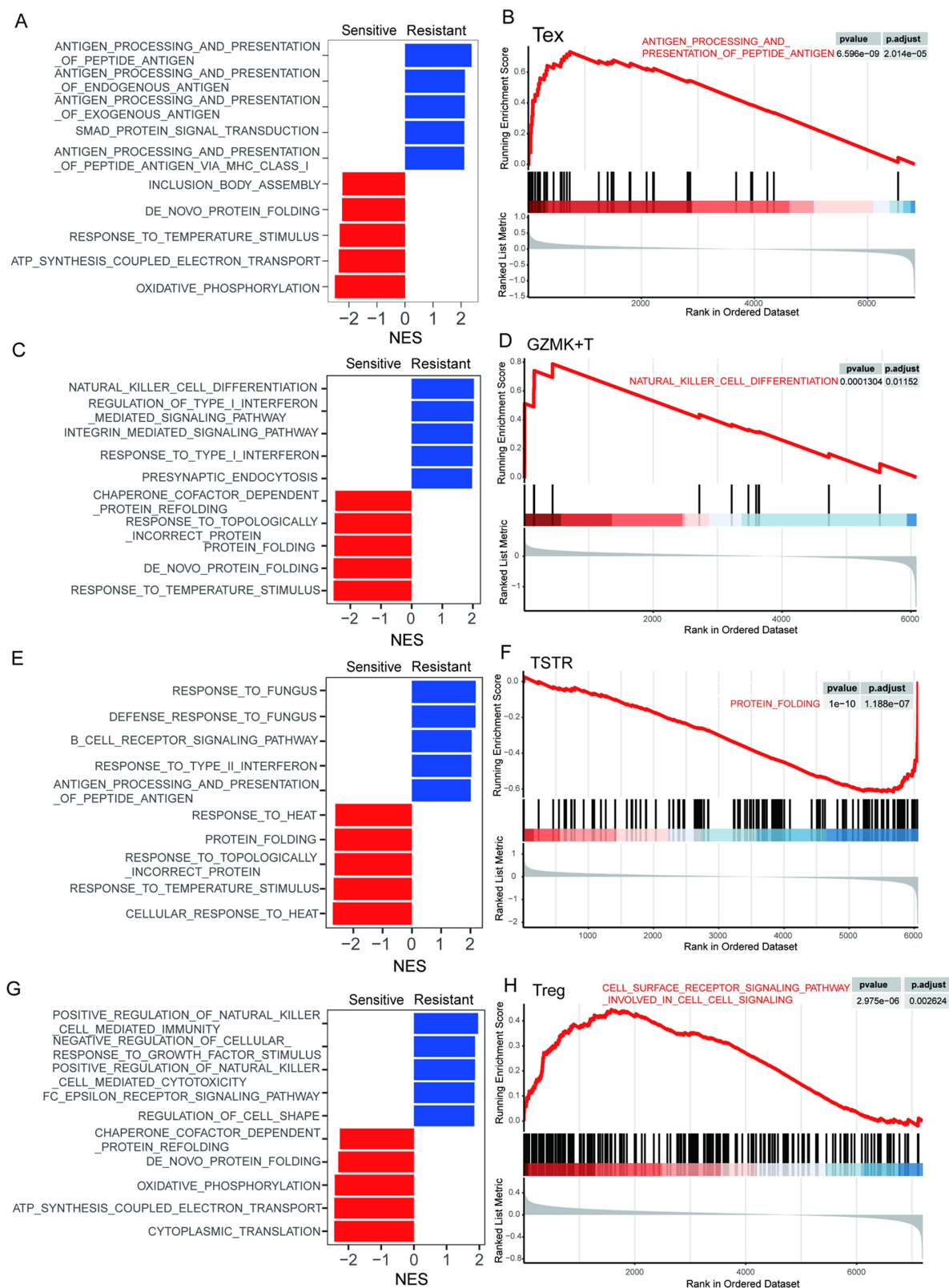


Figure 3. GSEA analysis identifying potential functional pathways driving tumor resistance in T-cell subsets. (A) GSEA analysis of Tex cells. (B) Antigen processing and presentation of peptide antigen pathways that were significantly enriched in the drug-resistant group in Tex cells. (C) GSEA analysis of GZMK+ T cells. (D) Demonstration of the natural killer cell differentiation pathway that was significantly enriched in GZMK+ T cells in the drug-resistant group. (E) GSEA analysis of TSTR cells. (F) Demonstration of the protein folding pathway that is significantly inhibited in TSTR cells in the drug-resistant group. (G) GSEA analysis of Treg cells. (H) Cell surface receptor signaling pathway involved in cell-cell signaling was significantly enriched in Treg cells in the drug-resistant group. GSEA, gene set enrichment analysis.

and presentation (Fig. 3A and B), whereas oxidative phosphorylation was markedly downregulated (Supplementary Digital Content Fig. S3A, available at: <http://links.lww.com/JS9/F57>) in the Tex cell-resistant group. Consistent with the observed suppression of Tex cells, GZMK+ T cells also exhibited reduced activity in response to oxidative or stress-related stimuli (Supplementary Digital Content Fig. S3B, available at: <http://links.lww.com/JS9/F57>). In contrast, GZMK+ T cells may enhance tumor resistance by activating NKT cells (Fig. 3C and D). Furthermore, GZMK+ T cell-mediated tumor resistance is characterized by the activation of the type I interferon signaling pathway and the suppression of the protein folding response (Fig. 3C). For TSTR cells, besides the significant activation of antigen processing and presentation pathways and suppression of heat- or temperature-related stress responses (Fig. 3E), the association with immunotherapy resistance appears to be more closely related to the downregulation of protein folding and molecular chaperone-mediated protein folding processes (Fig. 3F and Supplementary Digital Content Fig. S3C, available at: <http://links.lww.com/JS9/F57>).

In addition to CD8+ T cells, tumor cells can develop resistance to immunotherapy through the modulation of Treg recruitment. Our findings revealed a significant correlation between cell surface receptor signaling pathways and tumor-associated drug resistance in Treg (Fig. 3G and H). Furthermore, compromised intracellular ATP synthesis may contribute substantially to the development of tumor drug resistance (Fig. 3D). We observed that the proportion of unconventional T cells was significantly lower in the resistant group compared to the sensitive group, particularly in $\gamma\delta$ T cells (Fig. 2C and Supplementary Digital Content Fig. S2B, available at: <http://links.lww.com/JS9/F57>). This finding suggests that $\gamma\delta$ T cells play a pivotal role in anti-tumor immunity. Furthermore, GSEA revealed that $\gamma\delta$ T cells in the resistant group exhibited significant activation of the intra-tissue homeostatic pathway (Supplementary Digital Content Fig. S3E, available at: <http://links.lww.com/JS9/F57>). This highlights their role in maintaining intestinal tissue homeostasis and preserving barrier integrity. Importantly, we observed a significant activation of the protein folding pathway in the sensitive tumor group (Supplementary Digital Content Fig. S3F, available at: <http://links.lww.com/JS9/F57>), which further emphasizes the anti-tumor efficacy of $\gamma\delta$ T cells in immunotherapy.

Dysregulation of transcriptional regulatory networks plays a key role in the emergence of tumor resistance

TFs and their target genes in different T-cell subpopulations were identified using the SCENIC approach. We observed that Tex, GZMK+ T, TSTR, Treg, $\gamma\delta$ T, and other T-cell subpopulations exhibit distinct regulon expression patterns (Fig. 4A and Supplementary Digital Content Fig. S4A, available at: <http://links.lww.com/JS9/F57>). This indicates the functional diversity among T-cell subpopulations regarding their roles and specificity. A total of 22 TFs were found to be differentially expressed in TSTR cells. Among these, 20 TFs were significantly downregulated, while only 2 were upregulated in the drug-resistant group (Fig. 4B). Enrichment analysis of KEGG pathways for the potential target genes of these TFs revealed that the top-ranked TFs were primarily involved in signaling pathways such as MAPK, TNF, and IL-17 (Supplementary Digital Content Fig. S4B, available at: <http://links.lww.com/JS9/F57>). Notably, the differential

expression of KLF6, ATF3, FOS, and JUN is particularly significant (Fig. 4C), as these factors may influence numerous target genes (Fig. 4D). Additionally, we found that decreased FOS expression may have a significant impact on the prognosis of patients with MSI-H CRC (Fig. 4E). Similarly, we identified 24 TFs that exhibited significant differential expression in $\gamma\delta$ T cells (Fig. 4F), comprising 21 upregulated and 3 downregulated TFs. KEGG pathway analysis of the regulated target genes (Supplementary Digital Content Fig. S4C, available at: <http://links.lww.com/JS9/F57>) further indicated a significant downregulation of KLF6 and ATF3. Furthermore, NR3C1-regulated target genes were markedly enriched in pathways related to PD-L1 expression and the PD-1 immune checkpoint pathway in cancer (Supplementary Digital Content Fig. S4C, available at: <http://links.lww.com/JS9/F57>). These findings suggest that the significant downregulation of NR3C1 may be closely associated with drug resistance to anti-PD-1 therapy (Fig. 4G).

Interactions between T-cell subsets are critical for promoting tumor resistance and sensitivity

By comparing interactions between T-cell subpopulations, we observed a significant increase in both the number and strength of intercellular interactions in the resistant group relative to the sensitive group (Supplementary Digital Content Fig. S5A, available at: <http://links.lww.com/JS9/F57>). In the resistant group, cellular communication between various T-cell subpopulations was evident (Fig. 5A). Moreover, the analysis of afferent and efferent interactions demonstrated a significant enhancement in signal reception by both depleted and activated T cells, as well as increased signal transmission by TSTR, $\gamma\delta$ T, and other cells (Supplementary Digital Content Fig. S5B, available at: <http://links.lww.com/JS9/F57>).

Additionally, analysis of information flow ratios in key signaling pathways identified several signaling flows significantly enhanced or attenuated in the drug-resistant group (Supplementary Digital Content Fig. S5C, available at: <http://links.lww.com/JS9/F57>). Further ligand–receptor analysis indicated that MHC-I class signaling pathways were universally enhanced (Fig. 5B), suggesting a crucial role in T-cell activation. However, among the downregulated signaling pathways, CD69-KLRB1 was particularly noteworthy (Fig. 5C). Specifically, signaling from CD69, particularly its ligand expressed on TSTR cells, to KLRB1, the receptor for $\gamma\delta$ T, NK_T, and other cells, was significantly reduced in the resistant group compared to the sensitive group.

Screening of prognostic biomarkers in immunotherapy

Based on data from the Kaplan–Meier Plotter database^[29], the expression levels of six molecules (KLF6, ATF3, FOS, NR3C1, CD69, and KLRB1) were analyzed to investigate their potential association with the prognosis of patients receiving immunotherapy across multiple cancer types. High expression levels of FOS, CD69, and KLRB1 were found to be associated with improved overall survival (OS) and progression-free survival (PFS) in patients undergoing immunotherapy. In contrast, the expression levels of KLF6, ATF3, and NR3C1 were not significantly associated with patient prognosis (Supplementary Digital Content Fig. S6, available at: <http://links.lww.com/JS9/F57>). FOS and KLRB1 were selected for further investigation.

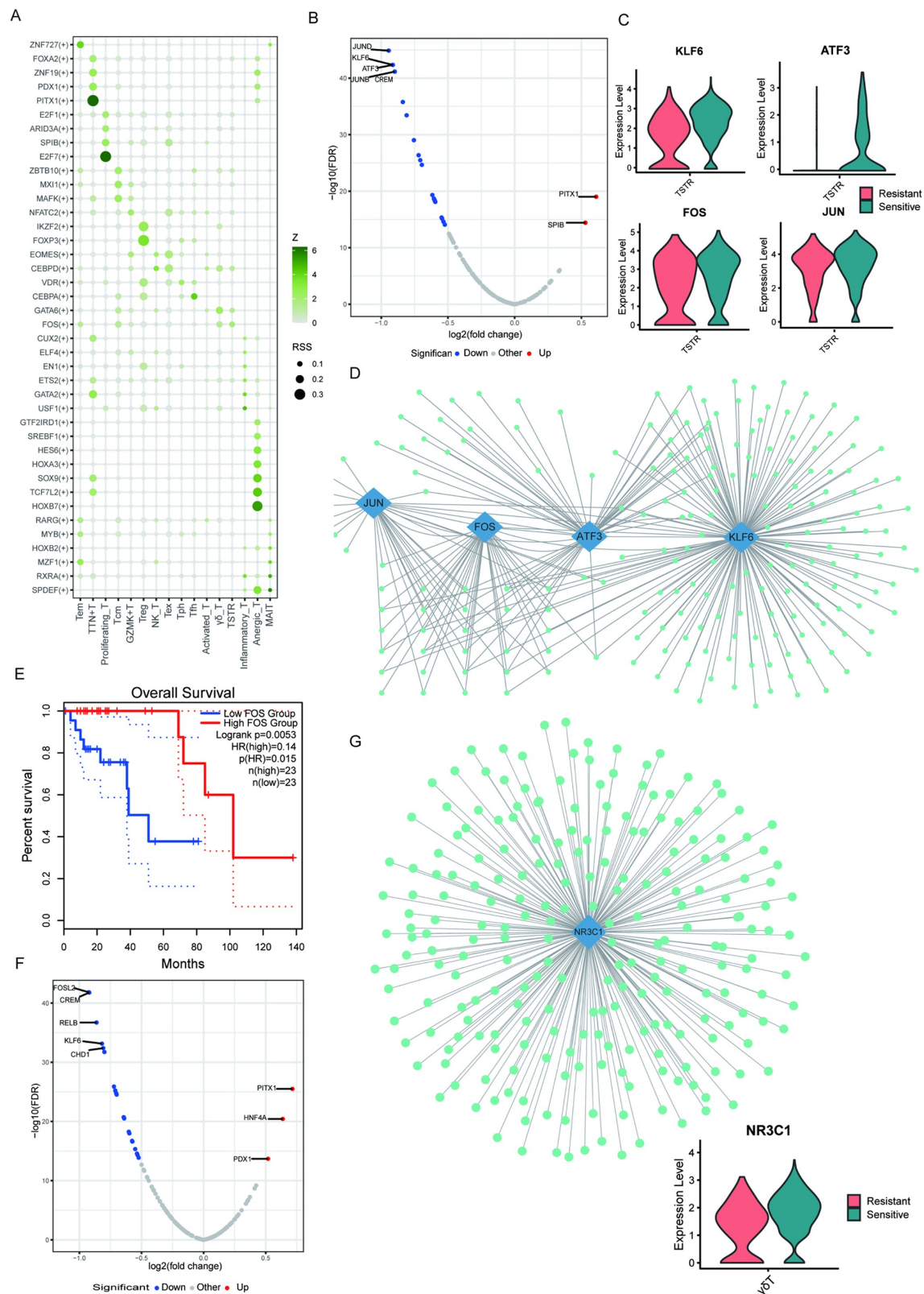


Figure 4. Dysregulation of transcriptional regulatory networks plays a key role in the emergence of tumor resistance. (A) Presentation of the RSS of regulons in each T-cell subset. (B) Differentially expressed TFs in the TSTR subset are illustrated. (C) Violin plot displays the expression levels of KLF6, ATF3, FOS, and JUN within the TSTR cell cluster. (D) Demonstration of the differential TF-mediated gene regulatory network in TSTR. (E) Prognostic outcomes of FOS based on GEPIA2 prediction in patients with MSI-H CRC. (F) Differentially expressed TFs in $\gamma\delta$ T are demonstrated. (G) Differentially TF-mediated gene regulatory networks in $\gamma\delta$ T cells, and a DEG violin plot between the two groups RSS, Regulon Specificity Score, TF, transcription factor; MSI-H, high microsatellite instability; CRC, colorectal cancer, DEG, differentially expressed gene.

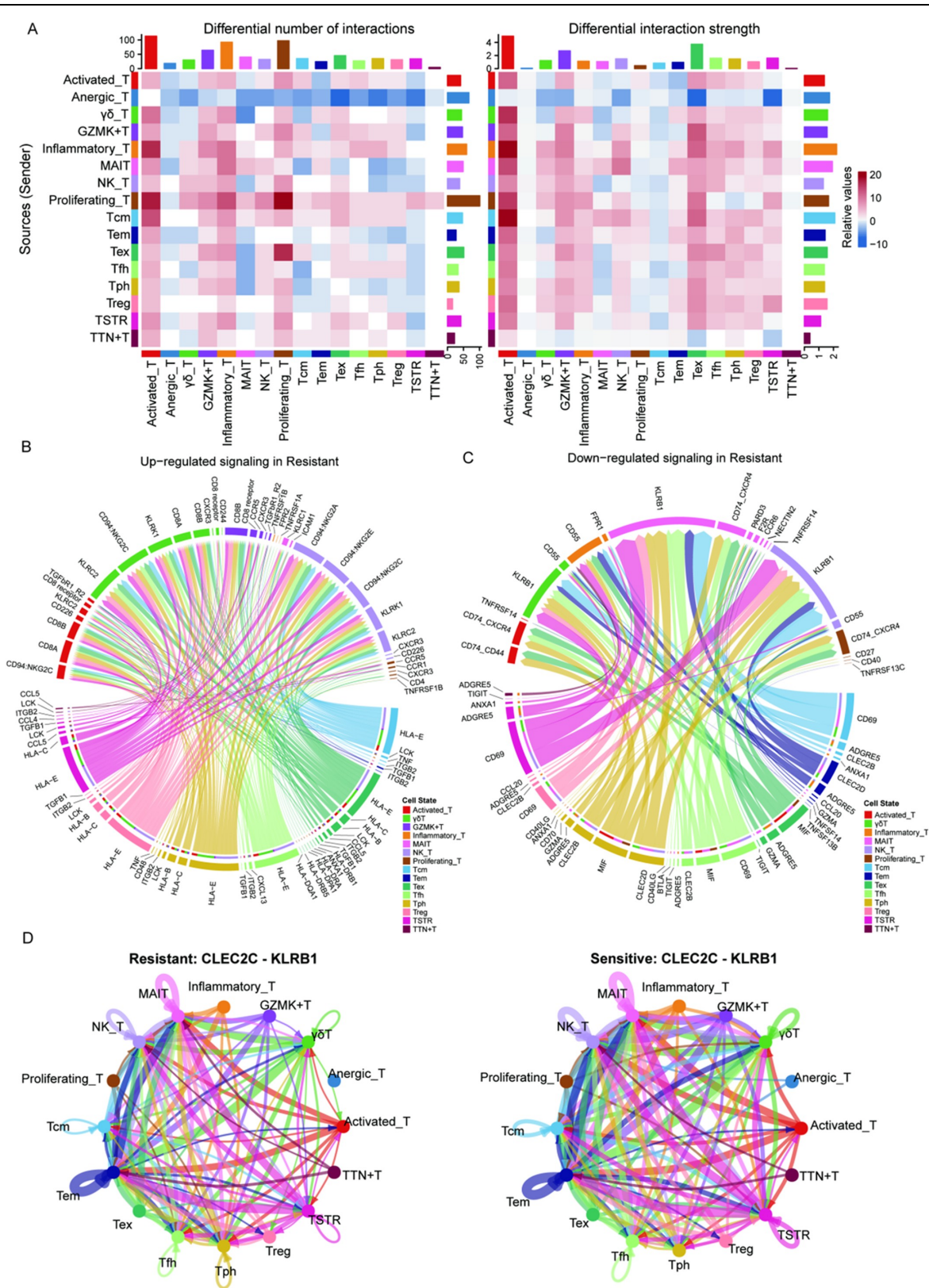


Figure 5. Interactions between T-cell subsets are critical for promoting tumor resistance and sensitivity. (A) Heatmap displaying the number and strength of interactions among different T-cell subpopulations. (B) and (C) Visualization of the upregulated and downregulated signaling ligand-receptor pairs in the resistant group relative to the sensitive group. (D) Comparison analysis of the CD69-KLRB1 signaling pathway in cellular communication between the resistant and sensitive groups.

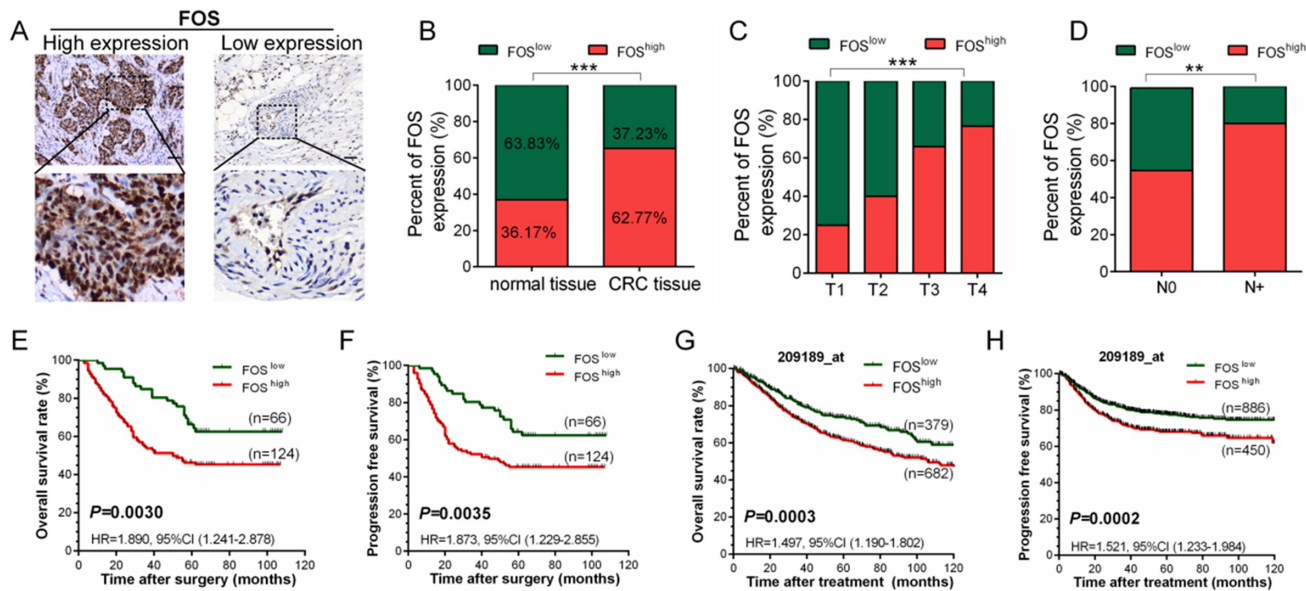


Figure 6. Expression of FOS in CRC and its association with clinical and pathological features. (A) Representative IHC staining images of FOS in CRC tissues and adjacent normal tissues. (B) FOS^{high} is significantly more prevalent in CRC tissues than in adjacent normal tissues. (C) FOS^{high} is positively associated with the depth of invasion. (D) FOS^{high} correlates with the presence of lymph node metastasis. (E) and (F) The Kaplan–Meier survival analysis revealed that shorter OS and PFS in patients with FOS^{high} CRC than in FOS^{low} patients. (G) and (H) Data from the Kaplan–Meier database confirmed that patients with FOS^{high} expression exhibited worse OS and PFS than those with FOS^{low} expression. CRC, colorectal cancer; IHC, immunohistochemistry; OS, overall survival; PFS, progression-free survival.

Expression of FOS and KLRB1 and their relationship with the prognosis of CRC patients

The expression level of FOS in cancer tissues was higher than that in the corresponding matched normal tissues (Fig. 6A). Moreover, the percentage of cases with FOS^{high} in CRC tissue (62.77%) was significantly higher than in matched normal tissue (36.17%) (Fig. 6B). Meanwhile, FOS^{high} was significantly associated with increased depth of invasion ($P < 0.001$) and the presence of lymph node metastasis ($P < 0.01$) (Fig. 6C and 6D). The relationship between FOS expression and clinical and pathological characteristics of CRC patients was subsequently analyzed. FOS protein levels were found to be significantly associated with tumor size ($P = 0.023$), invasion depth ($P = 0.012$), lymph node metastasis ($P < 0.000$), and TNM stage ($P < 0.000$) (Supplementary Digital Content Table S1-1, available at: <http://links.lww.com/JS9/F51>). Kaplan–Meier survival analysis of 190 CRC patients demonstrated that FOS^{high} was associated with significantly shorter OS ($P = 0.0030$) and PFS ($P = 0.0035$) compared to FOS^{low} (Fig. 6E and 6F). This finding was further validated using the Kaplan–Meier Plotter database (Fig. 6G and 6H). Univariate and multivariate Cox regression analyses demonstrated that FOS expression serves as an independent prognostic factor for both OS ($P = 0.047$) and PFS ($P = 0.045$) in patients with CRC (Supplementary Digital Content Tables S1-2, available at: <http://links.lww.com/JS9/F52> and S1-3, available at: <http://links.lww.com/JS9/F53>).

Subsequently, we evaluated the expression level of KLRB1 and its prognostic significance in CRC. IHC analysis demonstrated that the staining intensity of KLRB1 was markedly reduced in CRC tissues compared with that in matched adjacent normal mucosa (Fig. 7A). KLRB1^{high} was detected in 25.46 % of CRC specimens, compared to 53.75 % in adjacent normal mucosa

(Fig. 7B). Importantly, KLRB1^{high} was associated with a lower incidence of lymph node metastasis (Fig. 7C). Analysis of clinicopathological parameters revealed that KLRB1 protein expression was significantly associated with tumor depth ($P = 0.015$), lymph node metastasis status ($P < 0.001$), and TNM stage ($P = 0.002$) (Supplementary Digital Content Table S2-1, available at: <http://links.lww.com/JS9/F54>). Kaplan–Meier analysis revealed that KLRB1^{high} was significantly associated with prolonged OS ($P = 0.024$) and PFS ($P = 0.027$) compared to KLRB1^{low} (Fig. 7D and E). External validation using the Kaplan–Meier Plotter database supported these findings (Fig. 7F and G); MSI-H CRC patients with KLRB1^{high} demonstrated a more favorable prognosis compared to those with KLRB1^{low} (Fig. 7H). Consistently, serum CEA levels were found to be lowest in the cohort with KLRB1^{high} (Fig. 7I). Liver-specific MRI further illustrated differential immunotherapeutic responses in MSI-H CRC patients stratified by KLRB1 expression levels (Fig. 7J). Both univariate and multivariate Cox regression analyses confirmed that KLRB1 expression serves as an independent prognostic factor for OS ($P = 0.001$) and PFS ($P = 0.001$) in patients with CRC (Supplementary Digital Content Tables S2-2, available at: <http://links.lww.com/JS9/F55>, S2-3 available at: <http://links.lww.com/JS9/F56>).

Discussion

This project presents a comprehensive single-cell level analysis of anti-PD-1 monotherapy-resistant and -sensitive CRC tumor tissue samples. By analyzing the relative proportions of various cell populations, we found that T cells were the most predominant cell type and may play a critical role in immunotherapy.

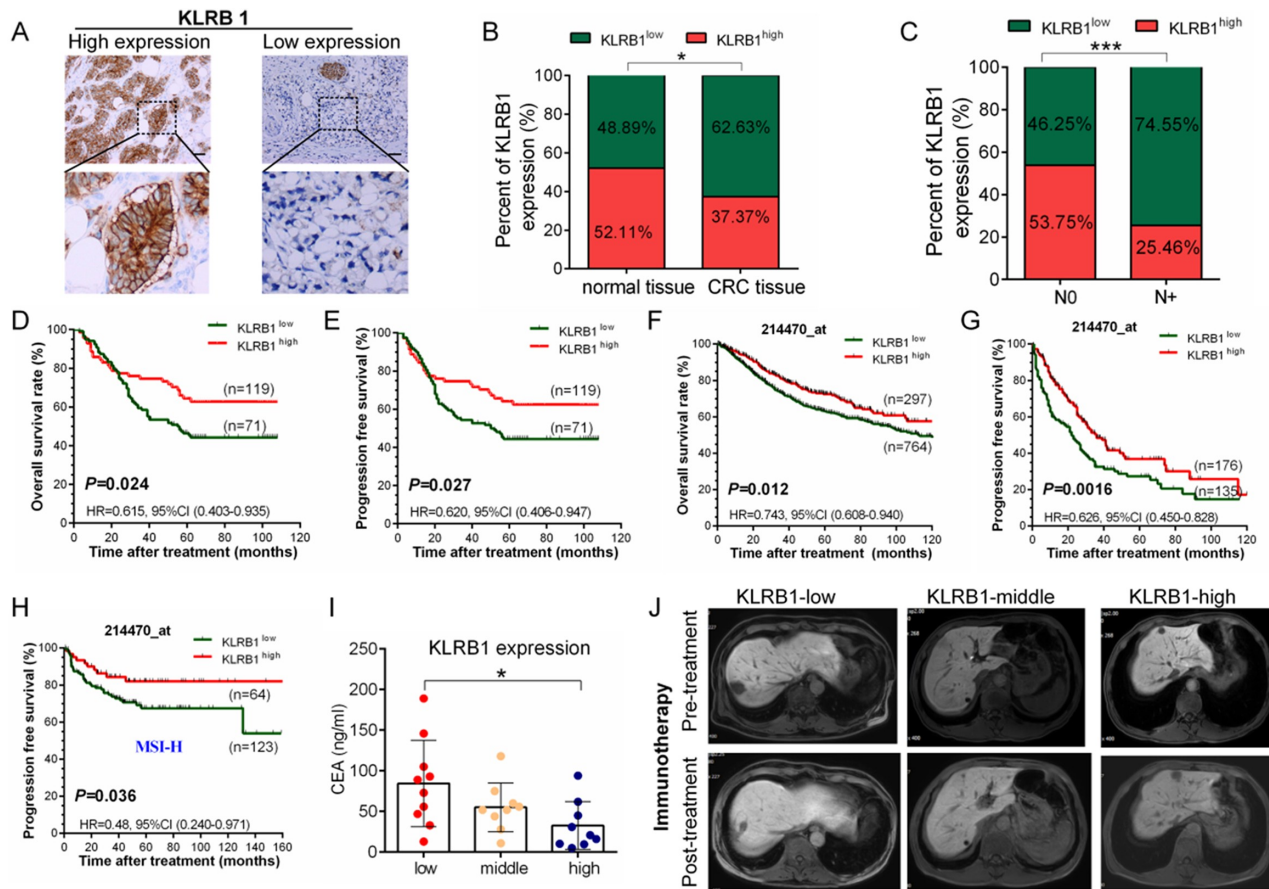


Figure 7. Expression of FOS in CRC and its relationship with clinical and pathological features. (A) Representative IHC images of KLRB1 in CRC tissues and adjacent normal tissues. (B) KLRB1^{high} is significantly lower in CRC tissues compared to the adjacent normal tissues. (C) KLRB1^{high} is negatively associated with lymph node metastasis. (D) and (E) Kaplan–Meier estimates survival analysis demonstrated that better OS and PFS in patients with KLRB1^{high} CRC than in KLRB1^{low} patients. (F) and (G) Data from the Kaplan–Meier database showed that patients with KLRB1^{high} exhibited a more favorable prognosis than those with KLRB1^{low}. (H) Data from the Kaplan–Meier database revealed that better PFS was observed in MSI-H patients with KLRB1^{high} than in those with KLRB1^{low} patients. (I) KLRB1^{high} is inversely correlated with CEA expression. (J) Changes in tumor volume as assessed by liver-specific magnetic resonance scans in MSI-H CRC patients with varying levels of KLRB1 expression after receiving immunotherapy. CRC, colorectal cancer; IHC, immunohistochemistry; MSI-H, high microsatellite instability; CEA, carcinoembryonic antigen; OS, overall survival; PFS, progression-free survival.

Furthermore, T cells can be subdivided into various subpopulations that exhibit distinct functional characteristics, and the relative proportions of these subpopulations differ markedly between the resistant and sensitive tumor groups. For example, T0 demonstrates elevated expression levels of specific inhibitory receptors, such as HAVCR2, LAG3, and TIGIT, and is frequently categorized as a Tex¹³⁰. Conversely, T5 is categorized as an intermediate state Tex (GZMK+ T) due to its elevated expression of GZMK, whereas T6 predominantly expresses heat shock proteins and other stress-related genes (HSPA1A, HSPA1B), which are associated with the TSTR¹³¹. Similarly, several additional subpopulations were identified, including activated CD8+ T cells expressing CX3CR1 (Activated_T), central memory T cells (Tcm, IL7R^{high}CCR7^{high}), effector memory T cells (Tem, IL7R^{high}CCR7^{low}), T follicular helper cells expressing CD4 (Tfh, CXCR5^{high}BCL6^{high}), peripheral helper T cells (Tph, CXCR5^{low}BCL6^{low}), and Treg co-expressing FOXP3 and CD25 (encoded by IL2RA). We investigated the roles of Tex, GZMK+ T, TSTR, Treg, and $\gamma\delta$ T cells in therapeutic resistance and identified several functional pathways that may be associated with drug resistance. TSTR cells are widely recognized

as being closely associated with tumor drug resistance¹³². In this study, TSTR cells were markedly decreased in the drug-resistant CRC tumor group, suggesting their potentially important role in mediating drug resistance in CRC. Additionally, an increase in Treg, which are key suppressor cells within the CD4+ T cells population, has been shown to contribute to the trigger of tumor drug resistance¹³³. With regard to unconventional T cells, a significant decrease in $\gamma\delta$ T cells was observed in the drug-resistant group, suggesting that these cells may exhibit substantial anti-tumor activity¹³⁴. Among other T-cell subsets, we observed an increase in the proportion of inflammatory T cells and a corresponding decrease in proliferative T cells within the resistant group. Additionally, we observed a significant increase in a T-cell subset characterized by high expression of TTN in the resistant group. This subset has been implicated in the T-cell inhibitory RhoA signaling pathway¹³⁵.

In summary, different T-cell subsets have specific functions and characteristics, which may contribute to exerting different immune effects in immunotherapy. Further regulatory subanalysis of TSTR and $\gamma\delta$ T cells indicated that dysregulation of transcriptional regulatory networks in the drug-resistant group

may contribute to the development of tumor drug resistance. TFs such as KLF6, ATF3, FOS, JUN, and NR3C1 are of interest owing to their potential roles in the regulation of PD-L1 expression or the PD-1 immune checkpoint pathway. These regulatory functions may affect the expression of downstream target genes, thereby contributing to the development of immune resistance. Antigen processing and presentation play a critical role in cancer immune evasion. Alterations in this pathway could result in impaired antitumor responses and therapeutic resistance^[36]. Oxidative phosphorylation represents a critical cellular metabolic process and has emerged as a target for effectively attenuating cancer drug resistance^[37]. Type I interferon signaling has been reported to contribute to the development of anti-PD-1 therapeutic resistance^[38]. Furthermore, TSTR cells are capable of influencing the protein folding process, which is mediated by molecular chaperones. When unfolded or misfolded proteins accumulate within the cell, they trigger the unfolded protein responses, which can interact with autophagy and may contribute to tumor drug resistance^[39].

Furthermore, in Treg, these findings provide valuable insights into the roles of different T-cell subsets in modulating immunotherapy responses. Abnormal activation or inactivation of cell surface receptor signaling pathways can lead to uncontrolled proliferation and impaired apoptosis of tumor cells, thereby increasing their resistance to treatment^[40]. Previous studies have demonstrated that intracellular ATP levels play a critical role in the emergence of drug resistance in colon cancer cells. Moreover, the direct delivery of ATP has been shown to induce resistance even in cells that were previously sensitive to drug treatment^[41]. Overall, the effects of Tex, GZMK+ T, TSTR, and Treg cells on tumor resistance in CRC are linked to the dysregulation of multiple functional signaling pathways. Current research into the immunotherapeutic potential of T cells mainly focuses on CD8+ and CD4+ T cells, which recognize antigenic peptides presented by MHC class I and MHC class II molecules, respectively. However, unconventional T cells that interact with MHC class Ib and non-classical MHC-I molecules have also been implicated in tumor immunity^[42]. The transcriptional regulatory network plays a crucial role in maintaining the normal function of the immune system. Dysregulation of various TFs has profound implications for tumor immune response and drug resistance. In this study, we conducted a comprehensive evaluation of the impact of TFs on T-cell subset functionality and their potential involvement in resistance to anti-PD-1 therapy. TSTR, representing a unique T-cell state, plays a crucial role in research concerning drug resistance following immune checkpoint blockade therapy. Concurrently, $\gamma\delta$ T cells are capable of directly recognizing tumor-associated antigens and rapidly eliminating tumor cells. These characteristics have made TSTR cells and $\gamma\delta$ T cells prominent subjects of research in the field of tumor immunotherapy. Consequently, we focused on investigating these two subpopulations within gene regulatory networks. We identified 22 TFs exhibiting significant differential expression. KEGG pathway enrichment analysis of the potential target genes regulated by these TFs revealed that the associated pathways are closely related to tumor drug resistance^[43–45].

KLF6 is recognized as a tumor suppressor and has been implicated in the development of chemoresistance in ovarian cancer. In contrast, ATF3 has been shown to regulate PD-L1 expression, which may enhance the effectiveness of anti-PD-1 immunotherapy^[46]. Furthermore, the expression levels of FOS

and JUN have been associated with the efficacy of anti-PD-1 therapy^[47]. The NR3C1 gene encodes the glucocorticoid receptor, and its downregulation has been associated with increased resistance to chemotherapy in colon cancer^[48].

These findings suggest that multiple functional TF-mediated gene regulatory networks are dysregulated in TSTR and $\gamma\delta$ T cells, which may contribute to tumor resistance following anti-PD-1 treatment. Overall, these results indicate that anti-PD-1 monotherapy significantly impacts the regulation of several immune checkpoint pathways in T cells. A more comprehensive understanding of the roles of different T-cell subsets in the immune response could facilitate the development of more effective immunotherapy strategies. Interactions among T-cell subpopulations play a crucial role in the immune system. These subpopulations collaborate to maintain immune homeostasis and defend against pathogen invasions through coordinated synergy and regulation. Similarly, interactions among T-cell subsets can have a substantial impact on the efficacy and resistance to immunotherapy. CD69 (CLEC2C) has been demonstrated to enhance the anti-tumor activity of immune cells in certain tumors and is regarded as a promising target for cancer immunotherapy^[49]. In contrast, KLRB1, which has been identified as a tumor suppressor, is associated with various immune characteristics of tumors. Moreover, the expression level of KLRB1 correlates with the sensitivity of cancer patients to chemotherapeutic agents. Therefore, KLRB1 may have significant potential as an immunotherapeutic target^[50]. We have identified a CD69-KLRB1 signaling pathway, the downregulation of which may play a critical role in driving drug resistance in CRC tumors. We hypothesize that the disruption of this signaling pathway could impair the function of T-cell subpopulations, such as $\gamma\delta$ T cells, which possess anti-tumor activity, thereby reducing their ability to recognize and attack tumor cells. Such disruption may allow tumor cells to escape immune surveillance, thereby potentially contributing to the development of drug resistance. These findings indicate that modulating the interactions among T-cell subsets could represent a strategy to counteract tumor resistance and enhance the effectiveness of immunotherapy.

Recent studies have highlighted the translational potential of scRNA-seq data in supporting clinical decision-making. For instance, a recent study characterized immune cell subpopulations in patients with non-small-cell lung cancer (NSCLC) and identified distinct T-cell signatures that are strongly predictive of responsiveness to immune checkpoint inhibitors^[51]. These findings highlight the clinical relevance of scRNA-seq as a platform for identifying predictive biomarkers, and our results in CRC further corroborate and extend upon this paradigm. Our findings suggest that the identification of immune suppressive T-cell subpopulations may serve as a predictive biomarker for patient response to immunotherapy. Specifically, T cells exhibiting elevated expression levels of inhibitory receptors, such as HAVCR2, LAG3, and TIGIT, may reflect the presence of immune resistance mechanisms, and targeting these pathways could potentially improve clinical outcomes. By integrating these biomarkers into clinical decision-making processes, clinicians may better select suitable candidates for immunotherapy, thereby potentially enhancing long-term outcomes. Using prospectively collected clinical specimens, we systematically investigated the associations between FOS and KLRB1 expression and clinicopathological parameters in CRC. FOS expression was significantly upregulated in CRC tissues, and higher FOS levels were significantly associated with poorer OS.

Multivariate Cox regression analysis further confirmed that FOS expression serves as an independent prognostic biomarker in patients with CRC. Previous studies have demonstrated that FOS regulates inflammatory CAFs, thereby enhancing tumor stemness, epithelial-mesenchymal transition, and hepatic metastasis via the SFRP1–FGFR2–HIF1 α signaling axis^[52]. Concurrently, activation of the MAPK cascade has been shown to drive CRC progression by promoting FOS phosphorylation; phosphorylated FOS subsequently trans-activates LINC02320 transcription, establishing a positive-feedback loop that amplifies oncogenic signaling^[53]. Collectively, these findings corroborate our observations; however, the precise molecular mechanisms by which FOS promotes CRC aggressiveness require further experimental investigation. KLRB1 expression is significantly downregulated in CRC tissues, and elevated KLRB1 levels are strongly correlated with improved OS. Multivariate analyses have confirmed that KLRB1 functions as an independent prognostic biomarker in CRC. In a prospective clinical trial, 42 patients with MSI-H CRC who received neoadjuvant PD-1 blockade were enrolled^[54]. Comprehensive genomic and transcriptomic profiling was conducted using next-generation sequencing, and immune-cell densities were quantified by multiplex immunofluorescence. The study revealed that CD8⁺ T cells exhibiting low PD-1 expression but high KLRB1 expression were significantly enriched in patients who achieved a pathological complete response. Feng *et al* further demonstrated that KLRB1 contributes to the formation of an immunosuppressive tumor microenvironment in CRC, modulates responses to PD-1 blockade, and is associated with hepatic metastasis^[55]. These independent observations provide supporting evidence for our findings.

Our findings suggest that immune-suppressive T-cell subpopulations may serve as reliable biomarkers for predicting patient response to immune-checkpoint blockade therapy. In particular, T cells exhibiting elevated expression of inhibitory receptors such as HAVCR2, LAG3, and TIGIT – are closely linked to mechanisms of immune evasion. Targeting these pathways through therapeutic blockade could potentially enhance anti-tumor immune responses. Integrating these biomarkers into clinical algorithms will facilitate the precision selection of patients for immunotherapy and is expected to improve long-term survival outcomes.

Conclusion

In summary, by analyzing the transcriptional landscape and intercellular communication networks of T-cell subsets in CRC, we demonstrate that dysregulated transcriptional control and altered cellular communication networks are key contributors to resistance against anti-PD-1 therapy. Among the DEGs, KLRB1 emerges as a promising predictive biomarker for stratifying patient response to immunotherapy. These integrative findings not only shed light on the molecular mechanisms underlying therapeutic resistance but also offer a rational foundation for the development of next-generation combination strategies aimed at overcoming adaptive immune evasion in CRC.

Ethics approval

This research has been approved by the Ethics Committee of the First Hospital of Lanzhou University (No. LDYYLL-2025-890).

Consent

All the patients signed informed consent for publication of this case report and accompanying images.

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Author contributions

Y.C., T.L., and D.L.: Research design; Y.C., G.M., and L.J.: Charts and data interpretation; Y.C., T.L., and G.M.: Literature search; C.W., D.X., and T.L.: Data collection and data analysis; Y.C., T.L., L.J.: Writing, editing; and all authors read and approved the final manuscript.

Conflicts of interest disclosure

The authors declare that they have no competing interests.

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Data availability statement

The authors will provide raw data on reasonable request.

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Statement about using artificial intelligence (AI)

During the preparation of this paper, the author utilized ChatGPT to improve the linguistic expression. Subsequently, we reviewed and revised the content. All authors affirm full responsibility for the content of the published article.

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